DAY-1

Git:

Git is a distributed version control system (DVSC) used to track changes in code & collaborate with others..

simple way:-

Think like this:

You are building a project

without git:-

- You will save file like
  - project_final_v1
  - project_final_v2
  - project_final_last_final..

Messing up & Not control

with git:-

- You track every change
- You can go back in time
- You can work safely without breaking things

Git is like = Google Docs history + undo system + Backup + collaboration tool combined

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Why Git Exists :-

Before git:-

- Developers overwrite each other's code
- No history tracking
- Hard to fix bugs

Git solves:-

- version tracking
- Collaborative
- Backup
- Safe experimentation.

Key concept {foundation}:

① Repository (Repo) ⇒ A project tracked by git

"It's an actual representation of your project"

② commit ⇒ A snapshot of your project at point in time

[An actual point when commit it will save what are the changes we have made].

③ Staging area ⇒ A place where you prepare changes before saving

④ Branching ⇒ A separate line of development..

Internal flow

working Directory {where you write code actual systemly

⇓

Staging Area {where you prepare changesly

⇓

Repository {where changes are permanently saved}

Example

You create file

app.py # directory Not tracked

git add app.py # Moving to stage..

Then

git commit -m " Added app.txt"

{ Save permanently }

So finally

" GIT is a distributed version control system that tracks changes in source code, allows multiple developers to collaborate, & maintain a history of all changes using commits"..